Ricardo N. Santos

PhD · Organic Geochemistry
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Organic geochemist, passionate about molecular paleoclimatology. I'm specialized in lipid biomarkers, compound-specific stable isotopes, and multi-proxy lake-sediment archives. My research focuses on disentangling hydroclimate, vegetation dynamics, and human–environment interactions from the Last Deglaciation to the present, with particular expertise in Central European and Mediterranean lake-sediment records.

Key Achievements

- **First-author in *Paleoceanography and Paleoclimatology*** (2026): Developed a precipitation δ^2H reconstruction based on vegetation-corrected plant-wax δ^2H values, revealing the influence of land-use change on molecular paleoclimate records.
- **First-author in *Quaternary Science Reviews*** (2024): Last 2000-year Atlantic-climate-vegetation reconstruction for western Iberia; 15 co-authors, multidisciplinary study.
- **EGU Climate Division Blog Editor** (2021–2025): Science communicator at Europe's largest geosciences society; Best Blog Post of the year by public vote winner.

Research Expertise

- Lipid biomarker and compound-specific isotope geochemistry
- Paleohydroclimate and vegetation reconstruction from lake-sediment archives
- Development and calibration of aquatic and terrestrial biomarker proxies
- Multi-proxy paleolimnology, organic geochemistry, and age–depth modelling
- Human–environment interactions in Holocene sedimentary records

Education

Ph.D. in Organic Geochemistry · University of Basel, Switzerland — *2022 – 2026*
Thesis: "Disentangling vegetation and hydroclimate signals from lipid biomarker records" ·
1st Supervisor: Prof. N. Ladd; 2nd Supervisor: Dominik Fleitmann

M.Sc. in Geology (Stratigraphy, Sedimentology, Paleontology) · University of Lisbon, Portugal — *2019 – 2021*
Thesis: "Hydroclimate of Western Iberia over the last 2000 years: Insights from leaf wax n-alkanes of Lake Peixão sediments (Serra da Estrela, Portugal)"

Postgraduate in Petroleum Geoscience · University of Lisbon — *2018 – 2019*

Postgraduate in Applied Geology · University of Lisbon — *2017 – 2018*

B.Sc. in Geology (Geology and Natural Resources) · University of Lisbon — *2013 – 2017*

Research Experience

Incoming Postdoctoral Researcher · Eawag, Swiss Federal Institute of Aquatic Science and Technology, Dübendorf, Switzerland — *Starting November 2026*

ERC-funded MEMELAND project: Investigating historical human–environment interactions through sedimentary fecal biomarkers and *sedaDNA*, organic geochemistry, and multiproxy lake-sediment records.

Doctoral Candidate & Project Assistant · Organic Geochemistry Group, University of Basel — *2022 – 2026*
Reconstructed climate–vegetation–human interactions across European and American lake-sediment records using lipid biomarkers, compound-specific isotopes, and multi-compound analyses.

ERASMUS+ Research Intern · Biología Costera Group, University of A Coruña, Spain
Sedimentary diatom analysis. — *Jun – Sep 2021*

Research Fellow · IPMA, Lisbon — *2021 – 2022*
Generated compound-specific δ^2H and $\delta^{13}C$ datasets from modern plant leaf waxes and lake sediments to investigate North Atlantic climate variability.

Research Fellow · FCiências.ID / Instituto Dom Luiz, Lisbon *2019 – 2021*
Integrated organic and inorganic lake-sediment proxies to reconstruct the influence of North Atlantic atmospheric circulation on western Iberian hydroclimate.

Exploration & Production Intern · Galp Energia, S.A., Lisbon *2017 – 2018*
Petroleum potential assessments: seismic interpretation, basin modelling, organic geochemistry data analysis.

PEER-REVIEWED PUBLICATIONS

[1] Fernández-Pérez, U., **Santos, R. N.**, Rodrigues, T., Schefuß, E., Bao, R., Raposeiro, P., et al. (**2026**). North Atlantic ocean–atmosphere dynamics controlled the hydroclimate in western Iberia during the Holocene. ***Paleoceanography and Paleoclimatology***. doi: 10.1029/2026PA005517

[2] Wangritthikraikul, K., Ladd, S. N., **Santos, R. N.**, Haghipour, N., Dubois, N., & Chawchai, S. **(2026)**. Multi-proxy reconstruction of Holocene paleoenvironmental changes in eastern Thailand: Sea-level and monsoon as key drivers. ***Quaternary Science Reviews***, 392, 110248. doi: 10.1016/j.quascirev.2026.110248

[3] **Santos, R. N.**, Nelson, D. B., Klatt, A., Schubert, C. J., Dubois, N., Jonge, C. D., & Ladd, S. N. **(2026)**. Central European Hydroclimate Since the Younger Dryas Inferred From Vegetation-Corrected Sedimentary Plant Wax δ^2H Values. ***Paleoceanography and Paleoclimatology***. doi: 10.1029/2025PA005401

[4] **Santos, R. N.**, Rodrigues, T., Naughton, F., Schefuß, E., Oliveira, D., Moreno, J., Raposeiro, P. M., Gil-Romera, G., Morgan, A., Leira, M., Gomes, S. D., Ladd, S. N., Trigo, R. M., Ramos, A. M., & Hernández, A. **(2024)**. Understanding the Atlantic influence on climate and vegetation dynamics in western Iberia over the last 2000 years. ***Quaternary Science Reviews***. doi: 10.1016/j.quascirev.2024.108796

[5] Costa, A. M., Freitas, M. D. C., Teixeira, A., Costa, P. J., **Santos, R. N.**, Cachão, M., Fatela, F., Pereira, R., Tereso, J. P., Lopes, V. P., Bao, R., & El-Boudjay, A. **(2024)**. Environmental changes and historical occupation at Oued Laksar (Ksar Seghir), Morocco. ***Quaternary International***. doi: 10.1016/j.quaint.2024.10.010

[6] Moreno, J., Ramos, A., Raposeiro, P. M., **Santos, R. N.**, Rodrigues, T., Naughton, F., Moreno, F., Trigo, R. M., Ibañez-Insa, J., Ludwig, P., Shi, X., & Hernández, A. **(2023)**. Identifying imprints of externally derived dust and halogens in the sedimentary record of an Iberian alpine lake for the past ~13,500 years – Lake Peixão, Serra da Estrela (Central Portugal). ***Science of The Total Environment***. doi: 10.1016/j.scitotenv.2023.166179

[7] Hernández, A., Sáez, A., **Santos, R. N.**, Rodrigues, T., Martin-Puertas, C., Gil-Romera, G., Abbott, M., Carballeira, R., Costa, P., Giralt, S., Gomes, S. D., Griffore, M., Ibañez-Insa, J., Leira, M., Moreno, J., Naughton, F., Oliveira, D., Raposeiro, P. M., Trigo, R. M., Vieira, G., Ramos, A. M. **(2023)**. The timing of the deglaciation in the Atlantic Iberian mountains: Insights from the stratigraphic analysis of a lake sequence in Serra da Estrela (Portugal). ***Earth Surface Processes and Landforms***. doi: 10.1002/esp.5536

[8] **Santos, R. N.**, Schefuß, E., Cordeiro, L. G., Oliveira, D., Hernández, A., Ramos, A. M., & Rodrigues, T. **(2022)**. Leaf wax biomarkers of a high-mountain lake area in western Iberia—Implications for environmental reconstructions. ***Frontiers in Environmental Science***. doi: 10.3389/fenvs.2022.994377

MANUSCRIPTS IN REVIEW/IN PREPARATION

[1] **R. N. Santos**, B. Aichner, D. B. Nelson, S. Hilt, S. N. Ladd. Lipid biomarker signatures of charophytes and submerged vascular macrophytes: Implications for past reconstructions. Accepted at: **Organic Geochemistry**

[2] Nannan Li, Dehui Li, Baojian Liu, **Ricardo N. Santos**, S. Nemiah Ladd, Daniel B. Nelson, Dorothy Sack, Ronald B. Lloren, Irene Brunner, Lisa C. Orme, Guizai Gao, Xinxin Zuo, Dongmei Jie, Nathalie Dubois. Proxy evidence challenges a humid Younger Dryas in Northeast China. (**in prep.**)

[3] Antonia Klatt, Fatemeh Ajallooeian, Petra Zahajská, **Ricardo N. Santos**, Daniel B. Nelson, Sebastian Naeher, Carsten J. Schubert, Nathalie Dubois, Cindy De Jonge, S. Nemiah Ladd. Natural and human-driven changes of phytoplankton community composition in Rotsee (Switzerland) since the Younger Dryas. Submitted to: **Quaternary Science Reviews**

[4] **R. N. Santos**, T. Wietelmann, D. B. Nelson, E. K. Thomas, C. A. Scholz, N. Scheuer, F. Naughton, F. Rey, N. Dubois, S. Szidat, S. N. Ladd. Vegetation-corrected plant wax δ^2H values from Owasco Lake record early Holocene hydroclimatic impacts of the residual Laurentide Ice Sheet. (**in prep.**)

ANALYTICAL AND TECHNICAL SKILLS

Lipid biomarker: ASE/Ultrasonication, saponification, derivatizations (methylation, silylation, acetylation), silica-gel and $AgNO_3$-silica column chromatography, urea adduction

Instrumentation: GC-FID, GC-MS, GC-IRMS for compound-specific $\delta^{13}C$ and δ^2H analysis.

Paleolimnology: Sediment core sub-sampling, bulk organic matter analyses, macrofossil picking, ^{14}C dating and $^{210}Pb/^{137}Cs$ geochronology, Bayesian age–depth modelling, diatom preparation, XRF core scanning.

Software: R, Grapher, QGIS/ArcGIS, Adobe Illustrator, PAST, Fiji/ImageJ

Field methods: Soil and vegetation sampling; Lake sediment coring (UWITEC), core description, stratigraphic logging

Languages: Portuguese (native), English (fluent), Spanish (fluent), Italian (Basic – in progress), German (Basic– in progress)

Selected Conference Contributions

IMOG – Porto (oral) *2025*

Reconstructing Past Precipitation δ^2H Values in a Small Swiss Lake: A Vegetation-Corrected Approach using Plant Wax Isotopes. **Ricardo N. Santos**, Antonia Klatt, Daniel B. Nelson, Carsten J. Schubert, Nathalie Dubois, Cindy De Jonge and Sarah N.Ladd

Urbino Summer School in Paleoclimatology – Urbino (oral & poster) *2025*

Reconstructing precipitation δ^2H values since the Younger Dryas in Central Switzerland: Correcting vegetation biases in plant-wax records. **Ricardo N. Santos**, Antonia Klatt, Daniel B. Nelson, Carsten J. Schubert, Nathalie Dubois, Cindy De Jonge and Sarah N.Ladd

EGU General Assembly – Vienna (oral) *2025*

*"*Disentangling the impact of vegetation and hydroclimate dynamics on plant wax stable isotopes since the Younger Dryas in Central Switzerland*"*. **Ricardo N. Santos**, Antonia Klatt, Daniel B. Nelson, Fatemeh Ajallooeian, Carsten J. Schubert, Nathalie Dubois, Cindy De Jonge and Sarah N.Ladd

EGU General Assembly – Vienna (poster) *2024*

"Comparison of lipid biomarkers and their hydrogen isotopic values in submerged vascular macrophytes and charophytes". **R. N. Santos**, B. Aichner, D. B. Nelson, S. Hilt, S. N. Ladd

22nd Swiss Geoscience Meeting – Basel (poster) *2024*

Impact of vegetation and hydroclimate dynamics on plant wax stable isotopes since the Younger Dryas in Central Switzerland. **Ricardo N. Santos**, Antonia Klatt, Daniel B. Nelson, Fatemeh Ajallooeian, Carsten J. Schubert, Nathalie Dubois, Cindy De Jonge and Sarah N.Ladd

Goldschmidt – Lyon (poster) *2023*

"Impact of Vegetation Dynamics on Leaf Wax Stable Isotopes Since the Younger Dryas in Central Switzerland". **Ricardo N. Santos**, Antonia Klatt, Daniel B. Nelson, Fatemeh Ajallooeian, Carsten J. Schubert, Nathalie Dubois, Cindy De Jonge and Sarah N.Ladd

Goldschmidt – Online (oral) *2022*

"Leaf waxes of a high-mountain ecosystem in Western Iberia as a tool for understanding past climatic and vegetation dynamics". **Ricardo N. Santos**, Enno Schefuß, Lívia Gebara Cordeiro, Dulce Oliveira, Armand Hernández, Alexandre Ramos and Teresa Rodrigues

EGU General Assembly – Online (oral) *2022*

"Climate and environmental changes over the last 2000 years in the Serra da Estrela, Portugal".

Santos, R. N., Rodrigues, T., Schefuß, E., Cordeiro, L., Naughton, F., Oliveira, D., Hernández, A. & Ramos, A.

EGU General Assembly – Online (oral) *2021*

"Hydroclimate reconstruction of Western Iberia over the last 3600 years - insights from lipid biomarker and specific isotope signal". **Santos, R. N.**, Rodrigues, T., Schefuß, E., Naughton, F., Oliveira, D., , Ramos, A., Hernández

Young Researchers in Geosciences Congress – Estremoz (oral & poster) *2019*

"Hydroclimate reconstruction of Western Iberia over the last 3600 years - insights from lipid biomarker and specific isotope signal". **Santos, R. N.,** P. J. M. Costa, I. Bosnic, A. Mena, A. N. Silva.

XI Iberian Congress of Geochemistry – Linares (oral) *2017*

"Characterization of REE(+/-Ba)-bearing Minerals from Granitic Rocks (Central Portugal) and Metassomatic Rocks (Bayan Obo, China)". **Santos, R. N.,** P. J. M. Costa, I. Bosnic, A. Mena, A. N. Silva.

Service, Outreach, & Awards

Peer Reviewer

- Quaternary Science Reviews
- *Palaeogeography, Palaeoclimatology, Palaeoecology*
- Organic Geochemistry

Co initiator, Younger Dryas Working Group *2025 – Present*
Coordinate an interdisciplinary working group bringing together lipid biomarker specialists, lake researchers, speleothem experts, glaciologists, chironomid specialists, and climate modelers to synthesize European proxy evidence for the Younger Dryas event and explore potential review-paper outputs.

BiG-CV *n*-alkyl lipid team contributor

Development of standardized biomarker nomenclature and synonym mapping.

Proposed Lead Convener — EGU General Assembly 2027 (provisional programme)

Spatio-temporal expression of cooling events of the last deglaciation and the early Holocene.

Proposed Co-Convener — EGU General Assembly 2027 (provisional programme)

Lipid biomarkers across the tree of life: novel molecular and isotopic tools for environmental and biogeochemical studies.

EGU25 Short-Course convener · "Radiocarbon Dating Concepts and Practical Guide"
EGU General Assembly Short Course · CL5/ESSI4 | [2025]
Short course on radiocarbon dating principles, AMS-based methods, sampling, calibration, reservoir corrections, and applications in environmental and paleoclimate research, including compound-specific ^{14}C analysis.

Blog Editor & Outreach Team · EGU Climate: Past, Present and Future *2021 – 2025*

- Contributed to science communication, blog editing, social media outreach, and early-career scientist activities.
- Supported the organization of outreach and networking events, including campfires and short courses.
- Winner of the **2022 EGU Best Blog Post Award – Public Vote** for *"A glimpse into the INTIMATE Summer School 2022."*

Media Outreach

- Research featured in **National Geographic Magazine Portugal** in the article *"O Tempo Preso na Lagoa" (2023)*.
- **Science communication video** · For Biogeochemistry and Sedimentology Labs of the *Portuguese Institute of the Sea and the Atmosphere (2021)*
  https://youtu.be/WWDSYlWuOYk

Scholarships & Competitive Support

- **Travel grant** from University of Basel to attend the Urbino Summer School in Paleoclimatology (**USSP 2025**); Urbino, Italy.
- **Travel grant** from Swiss Society for Quaternary Research to attend **Goldschmidt 2023**, Lyon, France, and present the poster *"Impact of vegetation dynamics on leaf wax stable isotopes since the Younger Dryas in central Switzerland."*
- **Travel grant** from PAGES to attend the **Numerical ecology and time series analysis of marine proxy data** (Copenhagen, Denmark) 2023 from Arctic Cryosphere Change and Coastal Marine Ecosystems led by Dr. Gavin Simpson.
- **Doctoral scholarship 2022 — FCT PhD, Fundação para a Ciência e a Tecnologia, Portugal**
  Awarded doctoral scholarship; ranked 2nd of 43 applicants in the Earth Sciences panel. Declined to pursue a PhD position at the University of Basel. Reference: 2022.11839.BD
- **Travel grant** from INTIMATE to attend the **INTIMATE-INQUA Summer School** 2022 (Turku, Finland)
  https://youtu.be/2F66bkFZ7iA
- **Postgraduate scholarship** from Petrobras International Braspetro B.V.
  Master's degree in Geology – University of Lisbon *2019 – 2020*
- **Postgraduate scholarship** from Petrobras International Braspetro B.V.,
  Postgraduation in Petroleum Geoscience – University of Lisbon *2018 – 2019*

Leadership & Civic Activities

- **Newsletter Editor**: Polar2E – College on Polar and Extreme Environments, University of Lisbon *2020 – 2022*
- **Treasurer:** Portuguese Association of Geoscientists and Petroleum Engineers *2018 – 2019*
- **President**: American Association of Petroleum Geologists - University of Lisbon Student Chapter *2017 – 2018*
- **Geological Team Coordinator**: Kosmonaus Association *2015 – 2022*
- **First Author**: Geology Blog: *Diário de um ProtoGeólogo* *2015 – 2017*